Learning and Individual Differences 113 (2024) 102470

Contents lists available at ScienceDirect

Learning and Individual Differences

journal homepage: www.elsevier.com/locate/lindif

Personalized gamification versus one-size-fits-all gamification in fully online learning: Effects on student motivational, behavioral and cognitive outcomes

Ya Xiao, Khe Foon Hew *

Faculty of Education, the University of Hong Kong, Pok Fu Lam, Hong Kong

ARTICLE INFO

Keywords:
Gamification
Individual differences
Personalization
Online learning

ABSTRACT

The "one-size-fits-all" approach commonly used in gamified learning is unlikely to optimize the effects of gamification due to individual differences. Personalized gamification based on player types has been considered a promising alternative. However, the jury is still out on whether personalized gamification improves student learning. Previous experiments examining its effectiveness have reported mixed results. This study employed a mixed-method research design to examine the effects of personalized gamification on student learning in terms of motivational (intrinsic motivation), behavioral (task completion rate), and cognitive (cognitive engagement and learning performance) outcomes, comparing it to a one-size-fits-all gamification approach over eight weeks. The results showed that students in the personalized gamification group ($n = 36$) significantly outperformed those in the one-size-fits-all group ($n = 29$) in motivational, behavioral, and cognitive outcomes. The follow-up individual interviews provide insights into the reasons underlying the reported effects.

Educational relevance and implications statement: The present study found that personalized gamification, tailored to individual differences in player types, has a significantly larger positive influence on student motivational, behavioral, and cognitive outcomes compared to the one-size-fits-all gamification approach widely used in education. Students' responses in the follow-up interview revealed that personalized gamification enhances their positive emotions and reduces negative emotions, which are antecedents of an improved motivational state. The improved motivational state exerts a positive influence on behaviors, which, in turn, promote cognitive learning outcomes. This study provides empirical evidence for the effectiveness of personalized gamification, sheds light on the underlying mechanisms of its effectiveness and offers guidance for its implementation in online learning.

1. Introduction

1.1. Gamification and learning

Gamification is the process of applying game elements to non-game contexts (Deterding et al., 2011). In the field of education, gamification is often referred to as gamified learning (Armstrong & Landers, 2017), where the learning process is modified to create engaging game-like experiences for learners (Landers et al., 2018). Another term used for educational purposes is game-based learning, which involves designing full-fledged (serious) games (Deterding et al., 2011). Gamification differs from game-based learning as it is a design process that incorporates game elements into learning to enhance its effectiveness, rather than creating a full game product.

Similar to the iterative game cycle (Garris et al., 2002), gamified learning activities are expected to motivate learners to continuously participate and progress through an engaging gamification loop (Liu et al., 2011). For example, the "goal-learn-reward" loop is a commonly used in gamified learning (e.g., Schöbel et al., 2023; Xiao & Hew, 2024). Learning goals can be presented as a list of tasks. Students demonstrate their accomplishment of goals by completing these tasks, before being rewarded by game elements such as points, badges, levels or virtual currency. Once rewarded, the loop restarts, encouraging learners to continue participating in the learning activity to earn more rewards. The same single game elements and constellations of those can be designed in diverse ways to maintain the loop. For instance, "Early Bird Badges" can be designed not visible to peers, functioning as goal signifiers to reminded learners that completing tasks before the due time was an

* Corresponding author at: Room 111A, Runme Shaw Building, The University of Hong Kong, Pokfulam Road, Hong Kong.
E-mail addresses: yaxiao@connect.hku.hk (Y. Xiao), kfhew@hku.hk (K.F. Hew).

https://doi.org/10.1016/j.lindif.2024.102470
Received 23 November 2023; Received in revised form 1 April 2024; Accepted 17 May 2024
Available online 27 May 2024
1041-6080/© 2024 Elsevier Inc. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

expected behavior goal (Huang et al., 2019). On the other hand, badges can be designed visible to peers, functioning as social markers to provide a social validation that the achievement activity is worthwhile (Kyewski & Krämer, 2018). Different gamification designs can have varying effects on their effectiveness. An absolute leaderboard helps intensify students' sense of comparison and competitiveness more than a relative leaderboard (Bai et al., 2021). The duration of the gamification loop in sustaining learners' interest is also a concern, as gamification may work through a novelty effect, where learners temporarily behave actively when exposed to a new or novel stimulus, but reduce that behavior when the novelty wears off (Sanchez et al., 2020).

1.2. Gamification and individual differences

Research scrutinizing the educational effectiveness of gamification has increased in the last decade and yielded inconsistent results. While meta-analyses have suggested its general positive impact (Bai et al., 2020; Sailer & Homner, 2020), some studies have reported its insignificant (e.g., Donnermann et al., 2021) or negative impact (Kwon & Özpolat, 2021). Thus, gamification is not effective per se (Sailer et al., 2017). Rather, effective gamification design requires a theoretical understanding of the motivational mechanisms through which gamification attains its impact (Krath et al., 2021).

Landers et al. (2018) theorized that game elements (predictors) affect motivation (motivational mediator), which, in turn, affects behavior (behavioral mediator), ultimately influencing cognitive learning outcomes. These causal pathways are moderated by personal or situational moderators. One key strand of moderator analysis concerns individual differences (Luo, 2022), whose moderating effect can be explained by the highly unique ways in which individual learners interpret game elements (Van Roy & Zaman, 2018).

According to the causality orientation theory, people's interpretations of the same event can differ based on their personal experiences. Some individuals may find an event intrinsically motivating, while others may perceive it as controlling (Deci & Ryan, 1985). The same principle applies to gamified learning, where the effects of game elements can differ among individuals. For example, while some individuals enjoyed the absolute leaderboard as it provided them with a sense of self-efficacy when ranked at the top, others at the bottom reported feeling the pressure of losing face (Bai et al., 2021). These different interpretations of the same game element may cause different motivation, behavioral and cognitive outcomes.

1.3. Personalized gamification

Most current gamification designs in education still employ the one-size-fits-all approach, which is unlikely to optimize student experience as this approach neglects the heterogeneity in learners' characteristics. Personalized gamification, whereby game elements are tailored to individuals' attributes, is extolled as an option to the one-size-fits-all approach (Rodrigues et al., 2023). Considerable efforts have been dedicated to identifying personal traits that can distinguish individual differences and contribute to the development of personalized gamified systems. Among these traits, player types based on gaming preferences have received the most extensive investigation (Oliveira et al., 2023). Player types are often utilized in game design to gain a better understanding of target users and customize the gaming experience according to their needs (Pessoa et al., 2024). Since gamification introduces game elements, understanding an individual's player type remains crucial and valuable for personalizing their gamification experience (Klock et al., 2020). In this study, we specifically investigate and apply player type models within the context of gamification in education, rather than focusing on a game setting.

1.3.1. Previous player type models

Commonly used player type models in exploring students' player types in gamified learning include Bartle's taxonomy (Bartle, 1996), Yee's gamer motivation profile (Yee et al., 2012), BrainHex (Nacke et al., 2014), and Hexad (Tondello et al., 2016). Bartle's (1996) taxonomy categorizes multiuser dungeons players into four player types: Achievers, who derive satisfaction from accumulating points and rising in levels; Explorers, who relish uncovering novel features and unraveling how things work; Socializers, who thrive on forming connections with other people; and Killers, who derive pleasure from killing off other players.

Drawing upon Bartle's model, Yee et al. (2012) identified three primary motivations in playing massively multiplayer online games: achievement, reflecting players' desire to overcome challenges and surpass others; socialization, highlighting players' inclination to interact with fellow gamers within the gaming realm; and immersion, capturing players' yearning to fully immerse oneself in the virtual game environment.

BrainHex is a gamer typology that draws on previous demographic game design models informed by personality typologies (Bateman et al., 2011; Bateman & Boon, 2006) and insights gleaned from neurobiological research (Nacke et al., 2014). BrainHex categorizes players into seven types: Achievers, deriving satisfaction from accomplishing goals; Conquerors, deriving satisfaction from triumphing over challenges and besting fellow players; Daredevils, seeking satisfaction through daring risk-taking; Masterminds, deriving satisfaction from solving complex problems; Seekers, deriving satisfaction from exploring new things; Socializers, deriving satisfaction from fostering connections and building relationships with other players; and Survivors, finding satisfaction in the thrill of terrifying experiences.

It is worthwhile to note that Bartle's taxonomy, Yee's gamer motivation model, and BrainHex were mainly devised to comprehend player types within the realm of gaming. When these models were applied to gamified learning, certain player types, such as killers, and survivors are usually not taken into consideration (e.g., Daghestani et al., 2020; Jurado & Meza, 2017) because fighting and thrills or terrors are not frequently employed in gamified learning scenarios. Additionally, learners may perceive game elements embedded into educational activities differently than they would in games (Tondello, 2019).

The Gamification User Types Hexad represents an expansion of player type from the realm of gaming to the domain of gamification (Marczewski, 2015; Tondello et al., 2016). This model, rooted in self-determination theory (SDT), classifies player types based on the inherent or extrinsic satisfaction derived from a task (Ryan & Deci, 2000). According to SDT, when a task fulfils three key psychological needs – competence (feeling capable of accomplishing tasks), autonomy (having control), and relatedness (feeling connected with others) – it becomes intrinsically enjoyable. If task completion leads to anticipated external outcomes, such as rewards, it becomes extrinsically satisfying. Additionally, providing a meaningful rationale enhances individuals' engagement in activities that may be uninteresting yet significant (Deci et al., 1994). The Hexad model identifies six player types: Achievers who enjoy completing challenging tasks (motivated by competence); Free Spirits who enjoy creating their own way and exploring within a system (motivated by autonomy); Socializers who enjoy interacting with others (motivated by relatedness); Philanthropists who enjoy altruistic actions without expecting anything in return (motivated by meaning); Players who enjoy earning rewards regardless of the activity type (motivated by extrinsic rewards); and Disruptors who enjoy disrupting the system either for negative or positive effects (motivated by the desire to trigger change). The category of Disruptors was not directly derived from SDT but rather observed in the online gamified systems.

1.3.2. The five archetypes

Considerable conceptual overlaps exist among the aforementioned established player type models, which may cause confusion and difficulty in selecting the right model to use within the field of gamified learning. Our goal in synthesizing player attributes is not to dismiss the

inherent overlap among them, but to propose a suitable player type model distinct enough to capture meaningful differences between players, yet concise enough for reasonable groupings of large numbers of players (Xiao et al., 2024). A parsimonious approach to achieve this is to synthesize the essential attributes from existing models (e.g., Bedwell et al., 2012; Hamari & Tuunanen, 2014).

To be specific, Achievers are present in all four previously mentioned player type models. Yee's definition of Achievers encompasses two key characteristics: being challenge-oriented and being competition-oriented, which aligns with the definition of Conquerors in the BrainHex model (Yee et al., 2012). BrainHex defines Achievers as more explicitly completion-oriented (Nacke et al., 2014). According to Hexad's definition, the completion of goals and the overcoming of challenges stem from the desire of self-advancement (Tondello et al., 2019). Therefore, two fundamental aspects can be identified across previous player type models: self-advancement and competition. As a result, two archetypes can be derived: Challengers, who are motivated by self-advancement through completing goals or overcoming challenges, and Competitors, who are motivated by competition with others.

Socializers are similarly defined across all four player type models, emphasizing the inclination towards social interactions with other players. In the context of gamified learning, social interactions particularly highlight the importance of collaborative and cooperative learning arrangements, where learners work in teams to achieve shared objectives (Sailer & Homner, 2020). In Hexad, Philanthropists exhibit significant overlap with Socializers. Upon closer examination of the way Philanthropists are operationalized, their attributes center around aiding others and sharing knowledge (Tondello et al., 2019). These aspects can be categorized under the collaborative and cooperative relationships highly valued by Socializers. Therefore, it is reasonable to derive the archetype Team Players, who are motivated by working together to accomplish shared objectives.

Explorers is "a confluence of interest mechanism and puzzle-solving" (Bateman & Nacke 2010, p. 4). BrainHex distinguishes these two foci by classifying Seekers as being motivated by the interest mechanism and Masterminds as being motivated by puzzle-solving. Free Spirits in the Hexad model are operationally defined as a combination of the interest mechanism and self-expression (Tondello et al., 2019). Yee et al. (2012) immersion is primarily loaded with exploring new things. The interest mechanism, at its core, represents curiosity towards new experiences. The puzzle-solving aspect focuses on tackling challenging problems, which aligns with the Challengers archetype. The self-expression element signifies students' desire for autonomy in performing tasks. Consequently, two distinct archetypes can be derived: Explorers, who are motivated by new things (e.g., surprises) and who desire to learn and understand new information, and Autonomy Seekers, who are motivated by autonomy in performing tasks.

As previously mentioned, player types like Killers, Survivors, and Daredevils were deemed unsuitable for the context of gamified learning due to the inappropriate nature of elements such as shooting, fighting, thrills, or terrors (Tondello, 2019). Additionally, Bartle's definition of Achievers is extrinsically motivated, which aligns with Players in the Hexad model. The Player type, as well as the Disruptor type, are not representative, as they only constitute a small proportion of the cohorts (Tondello et al., 2016). Consequently, these two player types are not considered in this study. Fig. 1 presents a visual representation of the synthesis of key player attributes aimed at achieving conceptual clarity.

Moreover, prior studies have reported suboptimal psychometric properties of previous player type models, which further highlights the need to reexamine the conceptualization of these models. Although BrainHex is one of the most frequently used questionnaires (Hughes et al., 2023), its reliability has been questioned by numerous studies. For example, Busch et al. (2016) discovered inadequate reliability of the BrainHex model, with Cronbach's α values for all player types except Socializer lower than 0.70. In another study, Monterrat et al. (2017) reported low internal consistency scores, ranging from 0.17 to 0.41, associated with the BrainHex classification. Similarly, Tondello et al. (2018) reported that all player types in BrainHex have low internal consistency (ranging from 0.22 to 0.52) using a large sample of 50,423. Typically, Cronbach's α values >0.70 are considered acceptable (Kline, 1999). With regard to Hexad, Santos et al. (2022) observed that the results did not support a well-fitted model with a Comparative Fit Index (CFI) of 0.70 and a Tucker-Lewis Index (TLI) of 0.67. Hexad was not well supported using an English sample (CFI = 0.72), German sample (CFI = 0.73), or Dutch sample (CFI = 0.77, TLI = 0.73) (Krath et al., 2021; Ooge et al., 2020). Typically, TLI and CFI values >0.90, along with RMSEA and SRMR values lower than 0.08, are indicative of an adequate model fit (Brown, 2015; Hu & Bentler, 1999). Given these circumstances, our proposed Pentagon player type model is expected to provide a better option for accurately measuring player types (see Appendices A and B for a comparison between our model and previous models).

1.3.3. Related work

Numerous studies have focused on modeling the relationships between player types and students' motivational and behavioral outcomes in gamified learning, revealing significant correlations (e.g., Bovermann & Bastiaens, 2020; Santos et al., 2021; Talavera-Mendoza et al., 2021; Ugur-Erdogmus & Çakır, 2022). In a limited number of studies that explored the effectiveness of personalized gamification through experiments, some have attempted to compare personalized gamification based on player types with a *non-gamified* approach (e.g., Challco et al., 2018; Challco et al., 2019; Lavoué et al., 2019) or *counter-personalized* approach which matches students with their least preferred game elements (e.g., Monterrat et al., 2017). However, these studies did not directly compare the effects of personalized gamification with the one-size-fits-all gamification approach that is widely used in gamified learning.

We only identified four studies that meet such criteria (Awais et al., 2019; Daghestani et al., 2020; Mora et al., 2018; Oliveira et al., 2020), indicating inconsistent results (see Table 1). In terms of motivational outcomes, a larger proportion of participants in the personalized gamification group expressed interest in having similar experiences in the future compared to the one-size-fits-all group (Mora et al., 2018). However, an insignificant difference was found in their flow experience (Oliveira et al., 2020). Regarding behavioral outcomes, more students in the personalized gamification group participated in self-assessment tasks compared to the one-size-fis-all group (Mora et al., 2018), and the weekly in-system time was also longer in the personalized gamification group than in the one-size-fits-all group (Awais et al., 2019). However, no significant difference was found in the number of submitted group tasks (Mora et al., 2018). In terms of cognitive outcomes, participants in the personalized gamification group achieved significantly more learning gains compared to those in the one-size-fits-all group (Daghestani et al., 2020). Similarly, it was found that a larger proportion of participants in the personalized gamification group achieved higher grades and a lower proportion of participants in the personalized gamification group achieved bad grades (Awais et al., 2019).

Overall, these four studies only focused on one or two aspects of learning, without conducting a comprehensive investigation of motivational, behavioral, and cognitive outcomes. Moreover, most of these studies provided no information regarding the experiment duration, or the duration was too short. For example, Oliveira et al. (2020) conducted an experiment that lasted only around 30 min. Additionally, none of these studies conducted follow-up surveys to gain a deeper understanding of participants' perceptions of the personalized gamification approach. Considering the aforementioned findings and limitations in previous studies, it is necessary to conduct more rigorous experiments to thoroughly examine the effectiveness of personalized gamification.

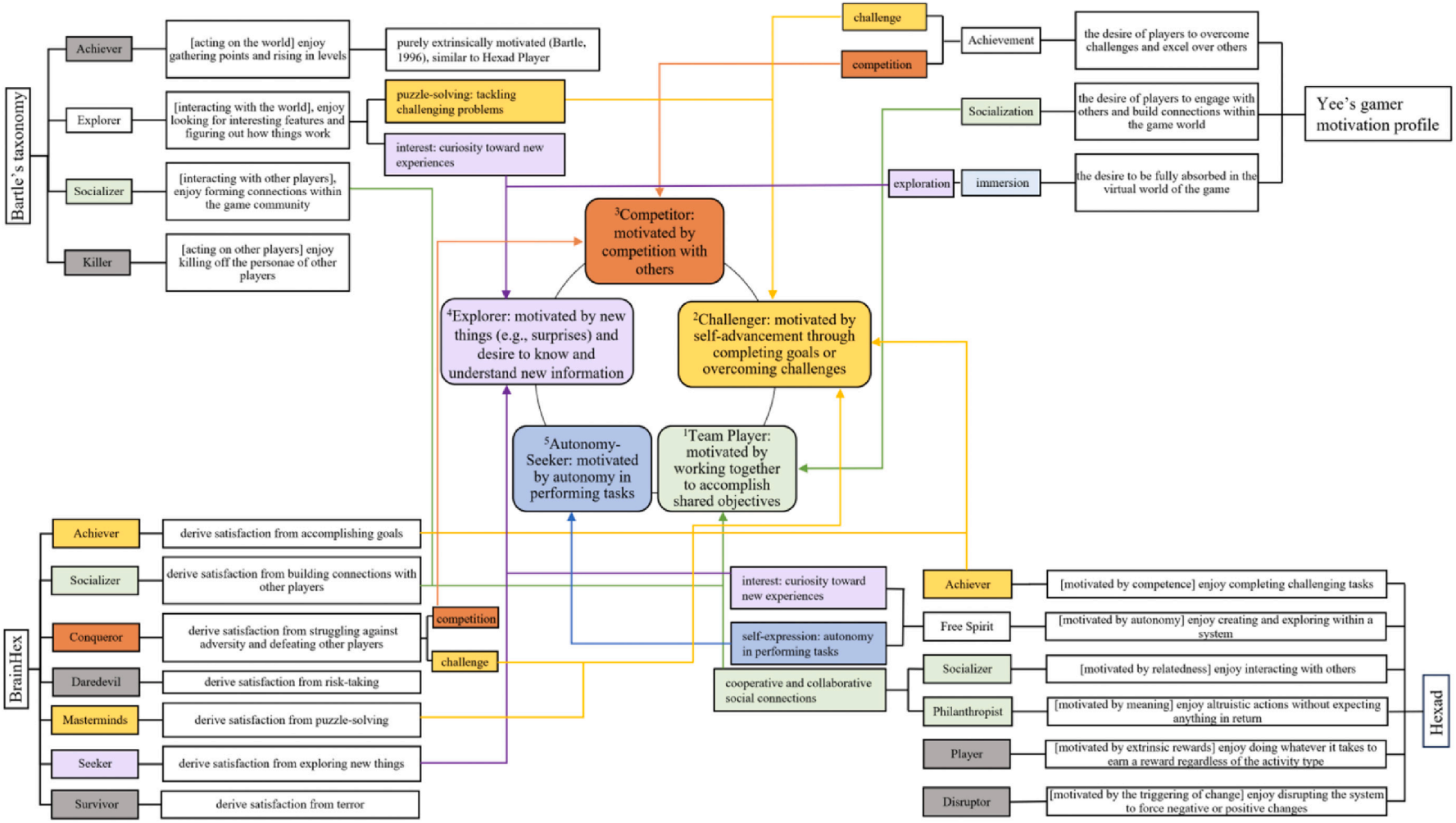


Fig. 1. Pentagon player type model – a synthesis of key player attributes in previous player type models

Table 1
Previous findings regarding the effectiveness of personalized gamification in comparison to the one-size-fits-all gamification.

Studies	Player type	Duration	Outcomes		
			Motivational	Behavioral	Cognitive
Mora et al. (2018)	Hexad	14 weeks	Interest – Positive (descriptive): A higher percentage of students agreed interest in having similar experiences in other subjects in the personalized gamification group than in the one-size-fits-all group (78.43 % vs. 52.63 %)	Participation – Positive (descriptive): The number of active students who participated in at least one self-assessment task in the personalized gamification group was more than that in the one-size-fits-all group, with 39 vs 11 Interaction – Insignificant ($p = 0.09$): The number of developed group programming tasks	/
Oliveira et al. (2020)	BrainHex	Around 30 min	Insignificant ($p > 0.05$): Flow experience		
Daghestani et al. (2020)	BrainHex	Not available			Positive ($p < 0.05$): significantly more learning gains between pre-test and post-test in the personalized gamification group than in the one-size-fits-all group.
Awais et al. (2019)	Social and self (self-summarized)	Not available		Positive (descriptive): The weekly average time spent in the system in the personalized gamification group was higher than that in the one-size-fits-all group, with 15 vs 10 h/w	A higher percentage of students getting good grades in the personalized gamification group than in the one-size-fits-all group, with 47.04 % vs 31.07 %; A lower percentage of students getting bad grades in the personalized gamification group than in the one-size-fits-all group, with 52.96 % vs 68.93 %

1.4. Contribution of the present study

This study makes the following contributions. First, we empirically investigate the effectiveness of personalized gamification by comparing the personalized gamification group and a well-designed one-size-fits-all gamification group over one whole semester. Second, we conducted a comprehensive investigation of student learning, operationalized as motivational, behavioral, and cognitive outcomes. Third, we used a mixed-methods research design. In addition to quantitative data, we examined students' perceptions to explain how/why personalized gamification works or doesn't work. Referring to the individual differences in perceptions of gamification, this study investigated the following research hypotheses.

Hypothesis 1. (H1): Participants in a personalized gamified learning condition indicate higher levels of motivational outcomes than participants in a one-size-fits-all gamified learning condition.

Hypothesis 2. (H2): Participants in a personalized gamified learning condition indicate higher levels of behavioral outcomes than participants in a one-size-fits-all gamified learning condition.

Hypothesis 3. (H3): Participants in a personalized gamified learning condition indicate higher levels of cognitive outcomes than participants in a one-size-fits-all gamified learning condition.

The interview data were used to address the following research question:

What are the students' perceptions of personalized gamification in fully online learning?

2. Methodology

2.1. Context and participants

The present study was conducted in a higher education context in Asia, involving 65 Asian participants with an average age of 21.52 ($SD = 1.56$). The participants were enrolled in a fully online international business course (course duration: 8 sessions, each lasting around 3 h). The class was selected using the convenience sampling method. Students signed up for this course voluntarily and expressed similar interest in it. A total of 36 students (80.56 % female, 19.44 % male) were assigned to the experimental group (EG), and 29 students (79.31 % female, 20.69 % male) were assigned to the control group (CG). There was no significant difference between CG ($M = 93.19$, $SD = 24.36$) and EG ($M = 100.73$, $SD = 20.46$) in terms of prior knowledge of the subject matter ($p = 0.21$, 95 % CI [−19.55, 4.46]).

To avoid possible bias introduced by different instructors, such as varying teaching styles, the same instructor taught both groups. The instructor was not informed which was the EG or CG and the researcher was not involved in teaching either group. Ethical approval for data collection from the Human Research Ethics Committee of the authors' institution and student consent were obtained in advance.

2.2. Gamification design

The course was delivered in a fully online mode using the flipped classroom approach, which consists of three major components: (1) asynchronous pre-class activities (e.g., watching video lectures and completing quizzes); (2) online synchronous in-class active learning (e.g., problem-solving); (3) asynchronous post-class assignments (Abeysekera & Dawson, 2015). All online in-class sessions were delivered via synchronous videoconferencing. The topic and content of each week were the same for both groups, so were the pre-, in- and post-class tasks. The course content and tasks were presented in the same sequence for the two groups.

In the asynchronous pre-class activity session, we prepared short instructional videos (Förster et al., 2022) and reading materials (Lai & Hwang, 2016) to assist students in recalling the necessary background knowledge and introducing basic concepts for each session's new learning topic. The students were then given five multiple-choice questions to test their retention of the pre-class materials. During the online synchronous in-class activities, the students focused on solving real-world problems related to business, such as conducting a market analysis for a company planning international expansion. At the end of each session, students were assessed through a short quiz comprising 10 multiple-choice questions, evaluating their understanding and recall of the learning content. Furthermore, after each online in-class session, students were given an essay question to evaluate their ability to apply the course content.

The points-adding rules were the same in both EG and CG. For each correct answer in the pre-class and in-class quizzes, a student was given 10 points, in which case, a maximum of 150 points could be earned for good performance in each session's quizzes. Five points were awarded for the first successful download of pre-class learning materials for each student, and five points were awarded for the completion of the pre-class tasks of downloading the pre-class materials and completing the pre-class quiz. Furthermore, the full mark for post-class assignments was 40 points. Once submitted, the teaching assistant gave personalized feedback.

2.2.1. Personalized gamification approach

For students in EG, the game elements were tailored to their player types. As shown in Fig. 1, we have proposed a new player type model that synthesizes essential player types in established player type models. This player type model comprises five main archetypes: Challenger, Competitor, Team Player, Explorer, and Autonomy Seeker. Building upon this model, we developed the Pentagon Player Type Questionnaire. To ensure the rigor of the Pentagon player type questionnaire, we employed a systematic development process containing five essential steps following previous guidelines (e.g., Artino et al., 2014; Neubauer et al., 2022; Rattray & Jones, 2007). Please see Table 2 for details. This questionnaire has been empirically validated by responses from 547 college students in the first round (Xiao et al., 2024) and another 360 college students in the second round. The full questionnaire is presented in Appendix C.

Based on students' player types identified through the Pentagon Player Type Questionnaire, the system provided corresponding preferred game elements. Specifically, for Challengers who are motivated by self-advancement through completing goals or overcoming challenges, the gamified activities were divided into three levels of difficulty: easy challenge (pre-class activities), medium challenge (in-class activities), and hard challenge (post-class activities). Students can progress through different levels by completing challenges of varying difficulty, with one level for easy challenges, three levels for medium challenges, and five levels for hard challenges.

For Competitors who enjoy competing with others, a leaderboard was presented, showing real-time rankings based on the students' accumulated points. At the end of each session, the leaderboard displayed a win statistic indicating the weekly top performers, which were the top three students.

Team Players who are motivated by working together to accomplish shared objectives were required to participate in the course in small groups. Each Team Player was randomly assigned to a group of 4 people, as small group size ranging from 2 to 6 has been shown to promote higher participation rates, which positively influenced learning performance in cooperative learning (Shaw, 2013). Their team assignment was to submit at least one problem they encountered or one strategy they used when solving the essay questions in the team discussion forum. Upon submission, they would be awarded an Excellent Team Member Badge. Additionally, they could monitor the task completion status of their team members.

For Explorers who are motivated by new things and have a desire to know and understand new information, they could unlock several links to extra learning sources (e.g., business news and magazines, case studies, etc.) in each session after completing certain tasks. Occasionally, they could also unlock hidden clues to solve the questions in the quizzes or the essay questions.

Autonomy Seekers who are motivated by autonomy in performing tasks were offered the chance to propose another essay question and submit their answers for personalized feedback in each session. They could also customize their settings by activating or deactivating game elements, even though the game elements matching their player types had already been initiated. Table 3 displays a detailed personalized gamification design in the flipped class.

Table 2
Five steps of the Pentagon player type questionnaire development.

Step	Purpose	Results
1. Conduct a literature review	• To reconcile the definitions of player types in previous player type models • To identify existing survey scales or items that might be used or adapted	• A new player type model comprising five main archetypes: Challenger, Competitor, Team Player, Explorer, and Autonomy Seeker • Four established scales were identified for reference: Motivations for Play in Online Games Scale (Yee et al., 2012), the BrainHex scale (Nacke et al., 2014), the Gamification User Type Hexad scale (Tondello et al., 2019). Additionally, an empirically validated survey measuring preferences for competition was referred to for the dimension of Competitor (Fallucchi et al., 2020).
2. Group the identified items into subthemes within each player type	To ensure that individual items tapped into the intended construct and comprehensively captured the core features of each dimension	The subthemes identified within each player type: • Challenger: overcoming obstacles, optimizing ability, completing goals, accumulating things, improving status, and achieving success • Competitor: exceeding a strong component, winning/being top-ranked, competitive, competition motivates one's best qualities • Team Player: conducting group activities, interacting with others (e.g., sharing knowledge), being part of a team • Explorer: fulfilling exploratory task, discovering hidden clues, curious, enjoying surprises • Autonomy Seeker: deciding for oneself, working out a meaningful solution, having self-expressions, independence
3. Develop items → Several iterations of discussions were held to revise inappropriate expressions until absolute agreement was reached on the final version of the questionnaire	To ensure items are clear, understandable and written in accordance with current best practices in survey design	An item pool containing 34 items was generated for the identified subthemes. Respondents were asked to indicate their agreement to the statements on a scale ranging from 1 (Very untrue of me) to 7 (Very true of me). Example items were:

(continued on next page)

Table 2 (*continued*)

Step	Purpose	Results
		• Challenger: It is important for me to keep challenging myself • Team players: I like cooperating with others • Competitors: I am a competitive person • Explorers: I like to have exploratory tasks (e.g., exploring external links related to the course topic). • Autonomy Seeker: I like to decide what to do rather than have the instructor tell me what to do
4. Conduct expert rating → An expert who had eight years of experience in researching gamification and two Ph. D. students majoring in psychometrics and educational psychology were invited to rate the items on a five-point Likert scale from 1 (not at all) to 5 (very much). They were asked to indicate to what degree they perceived each item to be clear (understandable) and valid concerning the definition of the player type.	To assess how clear and relevant the items are with respect to the construct of interest	Only items with a mean rating of 4 or more on this scale were retained. Seven items were removed in this step, resulting in a questionnaire containing 27 items.
5. Conduct factor analysis → A total of 547 Asian college students (28.34 % males and 71.66 % females) gave informed consent and filled out the questionnaire. The average age of participants was 21.32 years old (*SD* = 2.96). 82.27 % were undergraduate students, and 17.73 % were graduate students. Among them, 56.67 % major in Humanities and Social Sciences, 37.29 % major in Science and Technology, 5.85 % major in Medicine, 0.18 % major in others. → An additional sample containing 360 Asian college students (30.28 % male and 69.72 % female students) were collected. The average age of the participants was 22.79 years old (*SD* = 1.82). 96.94 % were undergraduate students, and 3.06 % were graduate students. Among them, 41.94 % major in Humanities and Social Sciences, 42.22 % major in Science and Technology, 10 % major in Medicine, 5.83 % major in others. → To examine factorial validity, confirmatory factor analysis was conducted using the maximum likelihood estimator with robust standard errors using the lavaan package (Rosseel, 2012) in R (R Core Team, 2022). → Reliability of each player type was measured by Cronbach's α.	To check for adequate validity and reliability of the questionnaire	• **Factorial validity**: **Criteria**: TLI and CFI values >0.90, RMSEA, and SRMR values lower than 0.08 are indicative of fair model fit (Brown, 2015; Hu & Bentler, 1999). **Results**: Three items were dropped as they were found contribute little information to the latent construct (i.e., player type) with small factor loadings (λ < 0.40). The revised model with 24 items showed satisfactory fit to the data according to all of the fit statistics. ***Using the 547 sample***: CFI = 0.91, TLI = 0.90, RMSEA = 0.07, 90 % CI = [0.06, 0.07], SRMR = 0.07. The correlations among the five player types were statistically significant, ranging from 0.11 to 0.68. A correlation <0.85 indicates that the subscales can be differentiated (e.g., Lee, 2019; Liu et al., 2020). The standardized factor loadings were also statistically significant and ranged from 0.50 to 0.88. ***Using the 360 sample***: CFI = 0.97, TLI = 0.96, RMSEA = 0.04, 90 % CI = [0.03, 0.05], SRMR = 0.05. The correlations among the five player types were statistically significant, ranging from 0.25 to 0.49. The standardized factor loadings were also statistically significant and ranged from 0.71 to 0.83. • **Reliability**: **Criteria**: Cronbach's α higher than 0.70 is considered acceptable (Kline, 1999). **Results**: ***Using the 547 sample***: The questionnaire as a whole showed excellent reliability with Cronbach's α standing at 0.92. Each of its five dimensions also had good reliability: Challenger (0.87), Team Player (0.89), Competitor (0.90), Explorer (0.77), and Autonomy Seeker (0.87). ***Using the 360 sample***: The questionnaire as a whole showed excellent reliability with Cronbach's α standing at 0.91. Each of its five dimensions also had good reliability: Challenger (0.90), Team Player (0.87), Competitor (0.90), Explorer (0.86), and Autonomy Seeker (0.88).

2.2.2. *The one-size-fits-all gamification approach*

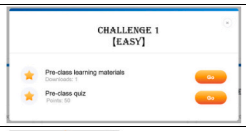
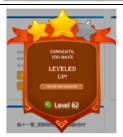
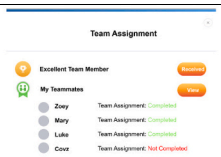
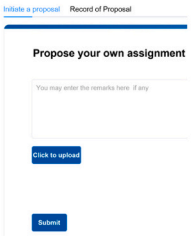
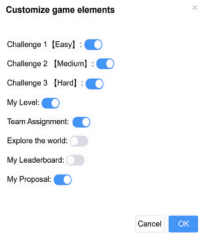
Students in CG were assigned to the one-size-fits-all condition featuring the points- badges-leaderboards triad. The points-badges-leaderboards triad has been frequently cited as one of the most employed sets of game elements in educational gamification research (Bai et al., 2020; Leitão et al., 2022). Therefore, it is considered a suitable one-size-fits-all approach (Rodrigues et al., 2023). The points-adding rules for the pre-class, in-class, and post-class learning activities remained the same as in the personalized gamification group. Moreover, students were awarded Task Completion and Task Master badges upon completing the pre- and in-class quizzes. Students who achieved an accuracy rate of higher than 80 % in the pre-class and in-class quizzes were awarded Fully Prepared and Quiz Whiz badges, respectively. Additionally, a real-time leaderboard was presented, showing the students' ranking based on their accumulated points. Table 4 shows the one-size-fits-all gamification design.

2.3. *Data collection and analysis*

2.3.1. *The profiling method of player types*

Previous studies that have utilized personalized gamification approaches have typically relied on the highest sum score of each player type to determine the dominant player type (e.g., Challco et al., 2018; Challco et al., 2019; Jurado & Meza, 2017; Oliveira et al., 2020).

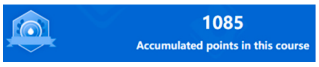
Table 3
Personalized gamification design.

Points-adding rules	Player types	Game elements	Example image
Pre-class: (1) 5 points for the first download of pre-class learning materials (2) 10 points for each correct answer in the pre-class quiz, 50 points in total (3) 5 points for the completion of the pre-class tasks In-class: (4) 10 points for each correct answer in the in-class quiz, 100 points in total Post-class: (5) Personalized feedback for the essay question, 40 points in total	Challenger	Challenge & Level: easy challenge (1 level) medium challenge (3 levels) hard challenge (5 levels)	T3.FIG1.TIF  T3.FIG2.TIF 
	Competitor	Leaderboard & Win statistic: leaderboard to show real-time rankings and win statistics of the top three performers. The score gap between you and the previous one is also updated in real time.	T3.FIG3.TIF
	Team Player	Social network: 4 people team to build connections, specifically, team forums to discuss problems or share strategies in solving the post-class essay question; access to the status of team assignments completed by each member	T3.FIG4.TIF T3.FIG5.TIF 
	Explorer	Unlockable & Exploration/navigation: external links for further exploration or hidden clues for solving problems conditioned on completion of certain tasks	T3.FIG6.TIF
	Autonomy Seeker	Customization: autonomy in proposing personal assignments and modifying the recommended game elements in the system	T3.FIG7.TIF 
			T3.FIG8.TIF 

Note. The full-size images are all available in a supplemental folder.
https://drive.google.com/drive/folders/13E2h9OyMhdNYXW5bnwNO8C050xlGpokP?usp=sharing.

However, this approach does not account for the possibility that a student may score equally or closely on multiple player types. To gain a more comprehensive understanding of player types, this study employed Cognitive Diagnosis Models (CDMs) (Xiao et al., 2024). CDMs are latent class models that classify individuals into latent classes based on their responses to assessment items or questionnaires, resulting in unified profiles of attributes (Shi et al., 2021). While CDMs are originally developed for cognitive diagnostic assessments to measure individuals'

Table 4
The one-size-fits-all gamification design.

Game elements	Example image
Points (The points-adding rules remained the same, see Table 3)	T4.FIG.1.TIF 
Badges (a) Task Completion badge for completing all pre-class tasks (b) Fully Prepared badge for students whose pre-class quiz accuracy >80 % (c) Task Master badge for completing the in-class quiz (d) Quiz Whiz badge for students whose accuracy rate in the in-class quiz was >80 %	T4.FIG. 2.TIF
Leaderboard Show the students' overall ranking based on their accumulated points	T4.FIG.3.TIF

Note. The full-size images are all available in a supplemental folder. https://drive.google.com/drive/folders/13E2h9OyMhdNYXW5bnwNO8C050xlGpokP?usp=sharing.

mastery profiles in specific skills, they have also been used to assess constructs such as personality traits (e.g., Revuelta et al., 2018) or psychological symptoms (e.g., Liang et al., 2023; Templin & Henson, 2006). In this study, the Pentagon player type questionnaire assessed five player attributes: Challenger, Team Player, Competitor, Explorer, and Autonomy Seeker. The resulting output of CDM analysis was the attribute profile for each participant, presented as either "1" or "0" for each attribute. For example, a profile of "11,010" would indicate that a participant exhibited attributes of Challenger, Team Player, and Explorer.

2.3.2. *The measures of student motivational, behavioral and cognitive outcomes*

Quantitatively, three sources of data were collected to analyze student motivational, behavioral, and cognitive outcomes in this study: (1) task completion rate, (2) test scores, and (3) questionnaire responses. Table 5 summarizes the methods used in the study.

To examine H1, students' motivational outcomes in the context of gamification were measured by questionnaire responses. We used the intrinsic motivation subscale from the Situational Motivation Scale (SIMS; Guay et al., 2000). The intrinsic motivation scale focuses on students' perceived interest and enjoyment in engaging in an activity. The Mann–Whitney *U* test was conducted to compare the CG's and EG's responses to the SIMS.

To examine H2, students' behavioral outcomes in gamified learning were measured by the task completion rate number of students who completed the pre-class, in-class and post-class tasks before the deadline. Chi-square tests were applied to test whether there was a statistically significant difference in task completion rates between the CG and EG.

In examining H3, students' cognitive outcomes in gamified learning comprised cognitive engagement (measured by questionnaire responses) and learning performance (measured by exam scores). The Motivated Strategies for Learning Questionnaire (MSLQ; Pintrich et al., 1991) was used to measure the students' cognitive engagement. The MSLQ was selected because it is commonly used to measure cognitive engagement (Greene, 2015), and it comprises the most comprehensive set of self-regulatory strategies specifically for undergraduate students, with robust scale reliability and validity (Broadbent, 2017; Pintrich et al., 1993). We chose the metacognitive self-regulation strategies subscale and the effort regulation subscale, as they fit the concepts of cognitive engagement explored in this study. Metacognitive self-regulation strategies involve students' ability to regulate and control cognition, including strategies such as goal-setting, planning, and self-monitoring. Effort regulation includes students' ability to control their effort in learning, reflecting their commitment to learning. The Mann-Whitney U test was conducted to compare the responses of CG and EG to the MSLQ.

The effects of personalized gamification on students' learning performance in gamified learning were measured by comparing their final exam scores. We administered a knowledge test to examine the initial differences in the students' relevant knowledge levels before the course started. At the end of the course, the students were given a final exam, and their scores were used to measure their learning performance. The full mark for the exam was 150. Analysis of Covariance (ANCOVA) was employed to compare the final exam scores of the CG with those of the EG after controlling for their prior knowledge.

Table 5
Measures of student motivational, behavioral and cognitive outcomes.

Student learning	Measurement	Data source	Analysis method
Motivational related (H1)	• Responses to the intrinsic motivation scale	SIMS (Guay et al., 2000)	Mann–Whitney U test
Behavioral related (H2)	• Completion rate of pre-class tasks • Completion rate of in-class tasks • Completion rate of post-class tasks	System log	Chi-square test
Cognitive related (H3)	• Responses to the metacognitive self-regulation strategies scale and the effort regulation scale	MSLQ (Pintrich et al., 1991)	Mann–Whitney U test
	• Final exam score	Final exam	ANCOVA

2.3.3. *The analysis of students' perceptions*

In addition to the above quantitative data, we also adopted a qualitative research approach by conducting individual interviews with students from the personalized gamification group within two weeks after the intervention. To ensure the likelihood of participation, we sent invitations to all 36 participants in the personalized gamification group and interviewed those who voluntarily agreed to participate.

Semi-structured interviews were adopted because their open-ended nature allows for flexibility (Huberman & Miles, 1994). Using this method, we were able to delve into students' perceptions of the personalized gamification approach in the fully online class. To guide the inquiry, we developed six main interview questions for each participant, as shown in Table 6.

In addition to the main interviewing questions listed, in practice, when discussing specific motivational, behavioral and cognitive outcomes, we followed up with prompts to explore the participants' concrete experiences. Examples of spontaneous follow-up questions included: "Can you give me examples to illustrate how personalized gamification motivates you to participate in this course?" and "Can you give me examples to illustrate how personalized gamification motivates you to self-regulate your study, for example, by setting goals and self-monitoring?"

Two individuals were involved in analyzing the interview qualitative data. Initially, the interview data were transcribed verbatim. The response data were first carefully reviewed to gain a holistic understanding. Next, the constant-comparative coding approach (Lincoln & Guba, 1985) was used to identify, analyze, and interpret patterns in

Table 6
Main interview questions.

Questions
1. The initial questionnaire reflects that you exhibit the attribute of X. Do you think this questionnaire captures your preferences accurately?
2. How do you feel about that we adopt personalized gamification in this course?
3. Do you think that the personalized gamification design motivated you to participate more in this course?
4. Do you think that the personalized gamification design motivated you to invest in more efforts to learn?
5. Do you think that the personalized gamification design motivated you to self-regulate your study more? If yes, how? If no, why not?
6. Do you have any suggestions for further improvements?

students' perceptions of personalized gamification, including its influence and reasons for such influence. The influence was guided by the main interview questions, which focused on motivational, behavioral and cognitive aspects. The reasons for such influence were not predetermined prior to our analysis but emerged and were continually refined throughout our interaction with the data. After the two coders had read and categorized the main themes, an inter-coder agreement of 96.25 % was achieved, indicating excellent intercoder reliability. Disagreements were resolved by discussion between the two coders until a mutual consensus was reached. In the following section, we provided exemplary quotes that effectively illustrated each main theme. The quotes were revised by removing unnecessary repetitions, slips, and other errors that do not contribute to the analysis. The quotes were initially in Chinese and then back-translated into English. Two bilingual individuals fluent in both Chinese and English were involved to cross-check the appropriateness of revision and accuracy of the translation (agreement = 94.38 %). The two individuals resolved disagreements by discussions. Please refer to the supplementary file for all quotes related to the main themes.

3. Results

3.1. Player type distribution in the personalized gamification group

Table 7 displayed the player type attribution of the 36 students assigned to the personalized gamification group. As shown, four participants exhibited single player attribute – Competitor. Eight participants exhibited two attributes: two exhibited the attribute of Challenger and Team Player; two exhibited the attribute of Challenger and Competitor; two exhibited the attribute of Challenger and Explorer; two exhibited the attribute of Team Player and Competitor; one exhibited the attribute of Team Player and Explorer. 17 exhibited three player attributes: Seven exhibited the attribute of Challenger, Competitor and Explorer; five exhibited the attribute of Challenger, Competitor and Autonomy Seeker; five exhibited the attribute of Challenger, Explorer and Autonomy Seeker. Six students exhibited four attributes: two exhibited the attribute of Challenger, Team Player, Competitor and Explorer; four exhibited the attribute of Challenger, Competitor, Explorer and Autonomy Seeker.

Table 7
Player type distribution.

Description	Number of participants
Competitor	4
Challenger & Team Player	2
Challenger & Competitor	2
Challenger & Explorer	1
Team Player & Competitor	2
Team Player & Explorer	1
Challenger & Team Player & Competitor	1
Challenger & Competitor & Explorer	7
Challenger & Competitor & Autonomy Seeker	5
Challenger & Explorer & Autonomy Seeker	5
Challenger & Team Player & Competitor & Explorer	2
Challenger & Competitor & Explorer & Autonomy Seeker	4

3.2. Motivational outcomes

The students' motivational outcomes were measured by the intrinsic motivation subscale from the SIMS, which displayed very good reliability in this study (Cronbach's alpha = 0.980). The response rate was 98 % (64 out of 65), with one student in EG did not respond to the questionnaire. The Shapiro–Wilk statistics showed a normal distribution of student responses in the CG ($W = 0.95$, $p = 0.15$), while it showed a non-normal distribution in the EG ($W = 0.64$, $p = 0.00$) in terms of intrinsic motivation. Accordingly, students' responses to this scale were examined using the Mann–Whitney U test. It was found that the students in the EG ($Mdn = 7.00$, $SD = 0.80$) had a significantly higher degree of intrinsic motivation than those in the CG ($Mdn = 5.00$, $SD = 1.03$, $U = 887.50$, $p < 0.01$, $r = 0.67$).

Thus, the results suggest that students given personalized game elements have a significantly higher level of intrinsic motivation compared to those who received the one-size-fits-all game elements.

3.3. Behavioral outcomes

3.3.1. Pre-class task completion

Fig. 2 presents the pre-class task completion information. As shown, the EG's completion rate was higher than that of the CG throughout all eight sessions. The completion rate of the EG held relatively steady and high, while that of the CG dropped drastically after Week 3. By the end of the course, 80.56 % of the students in EG had completed pre-class tasks while only 34.48 % of those in the CG had completed the pre-class tasks.

Table 8 displays the results of pre-class task completion. Considering the increased probability of Type I errors associated with conducting multiple Chi-square tests, the Sidak-Bonferroni procedure was employed to determine an adjusted alpha level of 0.006. It was found that although there was no significant difference in the pre-task completion rate in the first three weeks between EG and CG, students in EG indicated a significantly higher pre-class task completion rate than those in CG in

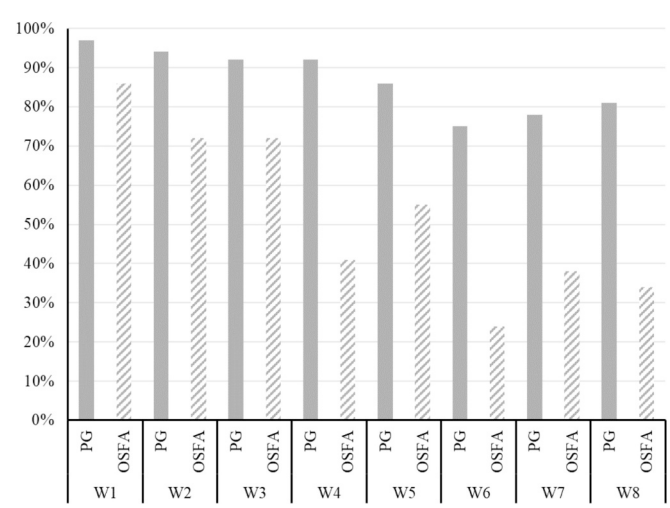


Fig. 2. Pre-class task completion rate before the deadline.

Table 8
Pre-class task completion.

Week	Condition	Completion	Chi-square
W1	EG	97.22 %	$X^2 = 2.75$, $p = 0.10$, 95 % CI [−0.03, 0.25]
	CG	86.20 %	
W2	EG	94.44 %	$X^2 = 5.99$, $p = 0.01$, 95 % CI [0.04, 0.40]
	CG	72.41 %	
W3	EG	91.67 %	$X^2 = 4.23$, $p = 0.04$, 95 % CI [0.01, 0.38]
	CG	72.41 %	
W4	EG	91.67 %	$X^2 = 19.07$, $p = 0.00$*, 95 % CI [0.30, 0.70]
	CG	41.38 %	
W5	EG	86.11 %	$X^2 = 7.68$, $p = 0.0056$*, 95 % CI [0.10, 0.52]
	CG	55.17 %	
W6	EG	77.78 %	$X^2 = 18.60$, $p = 0.00$*, 95 % CI [0.33, 0.74]
	CG	24.14 %	
W7	EG	77.78 %	$X^2 = 10.63$, $p = 0.00$*, 95 % CI [0.18, 0.62]
	CG	37.93 %	
W8	EG	80.56 %	$X^2 = 14.21$, $p = 0.00$*, 95 % CI [0.24, 0.68]
	CG	34.48 %	

* Significant using an adjusted-alpha of 0.006 (Bonferroni adjustment).

the subsequent five weeks.

3.3.2. In-class task completion

A higher proportion of students in the EG than in CG completed the in-class quizzes in all eight sessions of this course (see Fig. 3).

Chi-square tests of independence showed statistically significant differences between the EG and CG in terms of the completion rate of in-class quizzes in all eight sessions of the course (see Table 9).

3.3.3. Post-class task completion

Similar to the completion of in-class tasks, the descriptive data indicated that a higher proportion of students in the EG completed the post-class assignments relative to the CG in all eight sessions (see Fig. 4).

Chi-square tests of independence further showed that the discrepancy in post-class task completion rate was statistically significant in all seven sessions (see Table 10).

In summary, the results suggest that students who are provided with personalized gamified learning activities have a significantly higher level of behavioral engagement than those who went through the one-size-fits-all gamified activities.

3.4. Cognitive outcomes

This study explores two aspects of students' cognitive-related learning outcomes: cognitive engagement and learning performance.

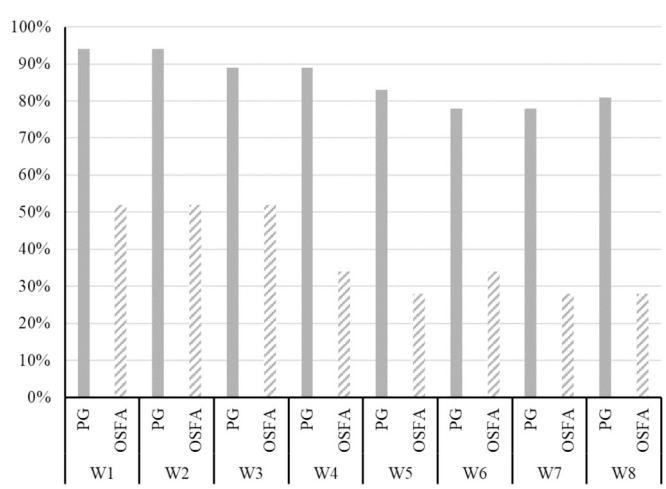


Fig. 3. In-class task completion rate before the deadline.

Table 9
In-class quiz completion.

Week	Condition	Completion	Chi-square
W1	EG	94.44 %	$X^2 = 15.80$, $p = 0.00$*, 95 % CI [0.23, 0.62]
	CG	51.72 %	
W2	EG	94.44 %	$X^2 = 15.80$, $p = 0.00$*, 95 % CI [0.23, 0.62]
	CG	51.72 %	
W3	EG	88.89 %	$X^2 = 11.08$, $p = 0.00$*, 95 % CI [0.16, 0.58]
	CG	51.72 %	
W4	EG	88.89 %	$X^2 = 20.79$, $p = 0.00$*, 95 % CI [0.34, 0.75]
	CG	34.48 %	
W5	EG	83.33 %	$X^2 = 20.56$, $p = 0.00$*, 95 % CI [0.35, 0.76]
	CG	27.59 %	
W6	EG	77.78 %	$X^2 = 12.40$, $p = 0.00$*, 95 % CI [0.21, 0.65]
	CG	34.48 %	
W7	EG	77.78 %	$X^2 = 16.38$, $p = 0.00$*, 95 % CI [0.29, 0.71]
	CG	27.59 %	
W8	EG	80.56 %	$X^2 = 18.38$, $p = 0.00$*, 95 % CI [0.32, 0.74]
	CG	27.59 %	

* Significant using an adjusted-alpha of 0.006 (Bonferroni adjustment).

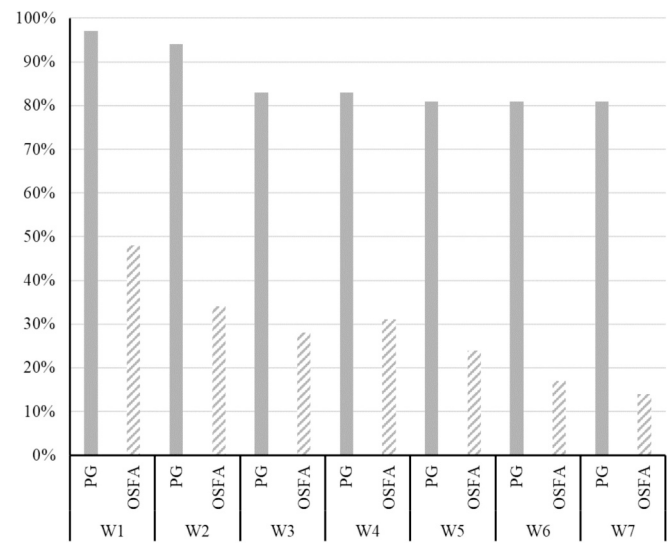


Fig. 4. Post-class task completion rate before the deadline. No post-class assignments were given on Week 8 since students were about to take the final exam.

Table 10
Post-class quiz completion.

Week	Condition	Completion	Chi-square
W1	EG	97.22 %	$X^2 = 20.74$, $p = 0.00$*, 95 % CI [0.30, 0.68]
	CG	48.28 %	
W2	EG	94.44 %	$X^2 = 26.41$, $p = 0.00$*, 95 % CI [0.41, 0.79]
	CG	34.48 %	
W3	EG	83.33 %	$X^2 = 20.56$, $p = 0.00$*, 95 % CI [0.35, 0.76]
	CG	27.59 %	
W4	EG	83.33 %	$X^2 = 18.31$, $p = 0.00$*, 95 % CI [0.32, 0.73]
	CG	31.03 %	
W5	EG	80.56 %	$X^2 = 20.69$, $p = 0.00$*, 95 % CI [0.36, 0.77]
	CG	24.14 %	
W6	EG	80.56 %	$X^2 = 25.81$, $p = 0.00$*, 95 % CI [0.44, 0.82]
	CG	17.24 %	
W7	EG	80.56 %	$X^2 = 28.64$, $p = 0.00$*, 95 % CI [0.49, 0.85]
	CG	13.79 %	

* Significant using an adjusted-alpha of 0.006 (Bonferroni adjustment).

3.4.1. Cognitive engagement

Students' cognitive engagement was also measured by the metacognitive self-regulation strategies subscale and the effort regulation subscale from the MSLQ. The two subscales indicated acceptable

reliabilities, with Cronbach's alpha = 0.80 and 0.72 respectively. As for the self-regulation strategies subscale, the Shapiro–Wilk statistics showed a normal distribution of student responses in CG ($W = 0.980$, $p = 0.85$) while a non-normal distribution of student responses in EG ($W = 0.81$, $p = 0.00$). Similarly, the Shapiro–Wilk statistics showed a normal distribution of student responses in CG ($W = 0.97$, $p = 0.49$) while a non-normal distribution of student responses in EG ($W = 0.88$, $p = 0.01$) in terms of the effort regulation strategies subscale. Accordingly, students' responses to these two scales were examined using Mann Whitney U tests.

The students in the EG ($Mdn = 6.60$, $SD = 0.73$) reported a significantly higher level of metacognitive self-regulation strategies than those in the CG, with a large effect size ($Mdn = 5.20$, $SD = 0.73$, $U = 870.50$, $p < 0.01$, $r = 0.62$). The students in the EG ($Mdn = 6.00$, $SD = 1.12$) reported a significantly higher level of effort regulation strategies than those in the CG, with a medium to large effect size ($Mdn = 4.75$, $SD = 0.10$, $U = 776.00$, $p < 0.01$, $r = 0.46$).

In summary, the results suggest that students who are provided with personalized gamified learning activities have a significantly higher level of cognitive engagement than those who went through the one-size-fits-all gamified activities.

3.4.2. Learning performance

Six students in the EG and three students in the CG did not participate in the final exam, thus our comparison of students' learning performance was based only on 56 students in total. Logistic regressions showed that the dropout of participants in the final exam was not related to their age, sex, intrinsic motivation, metacognitive self-regulation strategies and effort regulation strategies. Thus, we assume that the dropout of the six participants in the final exam was relatively random.

It was found that there was a significant effect of personalized gamification on students' performance in the final exam after controlling for their prior knowledge, with a medium to large effect size, $F(1, 54) = 5.46$, $p = 0.02$, partial $\eta^2 = 0.09$ (Cohen, 1969). Students in EG ($M = 103.13$, $SD = 22.39$) performed better than those in CG ($M = 87.23$, $SD = 22.00$) in the final exam. The assumptions of homogeneity of residuals were checked using the Modified Breusch-Pagan test ($p = 0.21$). The variances of the dependent variable was equal across groups according to Levene's test ($p = 0.98$). The homogeneity test of regression slopes was also checked ($p = 0.31$).

3.5. Perceptions of personalized gamification

We conducted individual interviews with the students in the personalized gamification group to explore their perceptions of the personalized gamification approach in the fully online class. 20 students participated. The main themes and related number of respondents are presented in Table 11.

Table 11
Qualitative (thematic) analyses of participants' reports of their perceptions.

Main themes	Number of respondents	Reasons
Boosted intrinsic motivation to learn	20 (100 %)	• Effective and efficient (19/20 = 95 %) • Feel being valued by the personalized design (15/20 = 75 %)
Motivated to participate in class	20 (100 %)	• Fun and interesting (15/20 = 75 %) • Less anxious (14/20 = 70 %)
Motivated to invest more efforts	19 (95 %)	• Less disorientation (11/20 = 55 %)
Motivated to regulate learning	18 (90 %)	
Suggestions for improvement	Static: 16 (80 %) Dynamic in the long run: 6 (30 %)	Player types may change over time, so the system should be improved to become adaptive and cater to these changes.

All interviewees (100 %) expressed that personalized gamification boosted their intrinsic motivation to learn. As they elaborated on the underlying reasons, five main factors emerged. Firstly, 95 % of interviewees reported that the personalized gamification design accurately targeted their preferences, making them perceive it as effective in improving learning. Example responses were:

> "I felt like a fish in its element. The personalized gamified learning activities suited me, so I felt more motivated to learn. This improved my learning efficiency."
>
> (Participant_3)

> "I liked the personalized design because it was targeted at everyone's preferences of gamified learning activities. I felt this was more effective in promoting learning progress."
>
> (Participant_8)

Secondly, 75 % of interviewees reported that the personalization aspect acknowledged and adapted to individual differences, giving them a sense of being valued. They appreciated the unique and meticulous design, which motivated them to put in the effort. Example quotes were:

> "This design provided different elements tailored to individual preferences, specifically making me feel this is a kind of customized service. I was pleased, so I paid more attention to learning."
>
> (Participant_1)

> "I felt my personal needs were being respected and valued. This gave me strength and confidence to learn."
>
> (Participant_18)

Thirdly, 75 % of interviewees reported that the personalized game elements were enjoyable and engaging, making the course less tedious and enhancing their enjoyment of learning. Example quotes were:

> "It stimulated my excitement, which enhanced my interest and participation in learning."
>
> (Participant_7)

> "The personalized gamified learning activities were quite interesting, making the course less boring. Therefore, I quite enjoyed it."
>
> (Participant_15)

Fourthly, 70 % of interviewees reported that in personalized gamification, anxiety-provoking game elements were filtered in advance. Without conducting a preliminary survey and adopting the personalized approach, negative feelings associated with certain game elements could have transferred to the learning experience. Example quotes were:

> "I preferred the personalized design. If I was given, let's say, a leaderboard, I might use it but felt stressed. Personally, I thought learning was my own business, so even if giving me some external pressure by comparing with others. It shouldn't have much effect on me."
>
> (Participant_5)

> "It's reasonable to have the personalized design. For example, when I saw the leaderboard showing the rankings of the accumulated points. I felt that I had to rank high. That gave me a sense of achievement. But for those who felt anxious about this, it's reasonable not to show this to them."
>
> (Participant_6)

> "Everyone has their own preferred way of learning. Without the preliminary survey and personalization, I might be reluctant to use those game elements that make me anxious. The reluctance towards the game elements might evolve into the reluctance to learn."
>
> (Participant_12)

Lastly, 55 % of interviewees reported that personalized gamification filtered out game elements that could potentially cause disorientation among students. If students felt lost in a gamified learning system, they

would be less likely to continue. Example quotes were:

> "I don't know what to do with game elements that are not suitable for me. I would be very upset and angry."
>
> (Participant_8)

> "If I was asked to do gamified activities that I couldn't handle, I would feel at a loss and wouldn't use the system."
>
> (Participant_11)

As a result of these positive perceptions and reduced negative emotions, all interviewees (100 %) reported increased participation in learning, such as improved attendance and task completion. For example, Participant_7 expressed:

> "I felt that the preliminary survey accurately catched my preferences. I liked the settings in general, so I tried to attend each live class and actively complete each task and develop a good habit of learning throughout the term."
>
> (Participate_7)

Additionally, 19 out of 20 interviewees (95 %) reported being motivated to invest more effort in learning than usual, particularly when dealing with laziness. Example quotes were:

> "This system motivated me not to give up easily when I was lazy. I would urge myself to work hard."
>
> (Participant_2)

> "I felt very good about this system. That's why I can stick to learning all the time and try to complete all tasks."
>
> (Participant_6)

Furthermore, 18 out of 20 interviewees (90 %) reported being motivated to regulate their learning. The personalized game elements served as goals throughout their learning. To achieve these goals, they monitored their understanding and made necessary adjustments, resulting in increased learning gains. Example quotes were:

> "For each session, my goal was to crack the third challenge. The third challenge was pretty difficult, which required me to apply what I learnt from the whole session. Therefore, I had to listen carefully and make sure I understood all the previous questions. The teachers' personalized feedback also made me reflect on my thinking. I could learn more solidly in this process."
>
> (Participant_7)

> "I particularly enjoyed checking links to extra learning sources, so I set weekly goals to actively complete required tasks and got access to the links. The links facilitated me to connect knowledge points."
>
> (Participant_12)

Finally, the interviewees provided suggestions for further improvement. Out of the 20 interviewees, the majority (80 %) reported that their player types remained stable during the intervention. However, six interviewees expressed their belief that their player types may change in the long run. As a result, they recommended implementing an adaptive method to accommodate such changes.

4. Discussions

In this study, we employed a mixed-methods research design to examine the effects of personalized gamification on student motivational (intrinsic motivation), behavioral (task completion rate) and cognitive learning outcomes (cognitive engagement and learning performance) in comparison to the one-size-fits-all gamification featuring the points-badges-leaderboards triad. It was found that students in the personalized gamification group significantly outperformed those in the one-size-fits-all group in terms of all three aspects of learning evaluated. The following sections revisit the major findings and discuss the theoretical explanations.

4.1. Positive effects on student intrinsic motivation (H1)

Students in the personalized gamification group reported significantly a higher level of intrinsic motivation than those in the one-size-fits-all group according to both the survey. This finding corroborates previous studies that have demonstrated the positive influence of personalization on student learning perceptions (Zhang et al., 2020; Zheng et al., 2022). The main reason for the positive effects can be attributed to students' enhanced pleasant emotions (e.g., enjoyment, gratitude) and reduced unpleasant emotions (e.g., boredom, anxiety) reported by students in their interviews.

These emotions are defined as achievement emotions – emotions that occur in response to events and actions that are judged according to competence-based standards of quality (e.g., studying) – in the control-value theory (Pekrun, 2006). Valence, which ranges from extremely pleasant to extremely unpleasant, is a key dimension of achievement emotions (Pekrun et al., 2023). When an activity is perceived as being sufficiently controllable by the self, pleasant emotions are assumed to be elicited. On the contrary, when an activity is perceived as lacking control, unpleasant emotions are experienced. For example, as reported by most interviewees, they enjoyed this course as the personalized game elements are interesting and fun to play with (enjoyment). They also felt being valued as the personalized gamification approach respects and accommodates student variability and preferences (gratitude). If without personalization, the given game elements may have no incentive value (positive/negative) to them, so they might be at a loss (disorientation). Some students expressed anxiety about competition if forced to participate using leaderboards, which might lead to ineffectiveness in promoting learning progress (anxiety). The valence of achievement emotions plays a crucial role in shaping motivational consequences. Pleasant emotions are assumed to strengthen intrinsic motivation, while unpleasant emotions are held to be detrimental to intrinsic motivation (Pekrun et al., 2023).

4.2. Positive effects on student behavioral and cognitive outcomes (H2 & H3)

Students in the personalized gamification group showed significantly higher levels of behavioral and cognitive outcomes than those in the one-size-fits-all group. Specifically, the pre-, in-, and post-class task completion rates were significantly higher in the personalized gamification group than in the one-size-fits-all group. Students in the personalized gamification group reported significantly higher levels of cognitive engagement measured by the ability to use self-regulation strategies and control efforts. Moreover, students in the personalized gamification group achieved significantly better performance in the final exam than those in the one-size-fits-all group. These findings are in line with previous experiments that found positive effects of personalization on students' behavioral engagement (e.g., Yang & Ogata, 2023), self-regulation (e.g., Zheng et al., 2021), and learning performance (Zheng et al., 2022). These positive relationships can be attributed to the linkage between motivational mediators and behavioral mediators as well as cognitive outcomes.

Several fundamental theories can shed light on the crucial role of intrinsic motivation in human behavioral change, such as the social cognitive theory (Bandura, 1977) and the self-determination theory (Ryan & Deci, 2000). As mentioned earlier, emotional arousal is a constituent source of intrinsic motivation as it can impact people's self-efficacy. When individuals are not beset by aversive arousal triggered by unpleasant stimuli, such as anxiety caused by leaderboards, they are more likely to expect success. Therefore, individuals tend to take actions to avoid unpleasant situations, such as skipping class. On the other hand, the satisfaction of intrinsic motivation derived from environmental stimuli enhances positive behavioral change. The theory of gamified learning also emphasizes the causal relationship between motivation state and behavior change in the gamified learning context. (Landers,

2014; Landers et al., 2018). Many interviewees reported that they felt motivated them to complete class tasks on time because the personalized gamification design in the class was inherently satisfying, providing direct empirical evidence for this relationship.

Additionally, Pekrun et al. (2023) examined the direct link between emotions and the use of cognitive strategies, effort investment and self-regulation. It was found that enjoyment was positively related to measured cognitive learning outcomes, while all negative emotions were negatively related to self-regulation. Furthermore, positive emotions were significant positive predictors of learning performance, whereas negative emotions were significantly negatively related to learning performance. The theory of gamified learning extends this link by illustrating that the positive effects of personalized gamification on cognitive outcomes are causally contingent on motivational and/or behavioral changes within an individual (Landers et al., 2018). According to the interviews, the personalized gamification design motivated students to be more concentrated and persistent, even when the course contents were boring. Some of them were motivated to set weekly goals and make plans for reaching these goals. As a result, they felt that they achieved greater learning gains.

4.3. Students' perceptions of personalized gamification in fully online learning

Students' perceptions towards the personalized gamification approach were generally positive, with 100 % reporting that this approach promoted their intrinsic motivation to learn, 100 % reporting that it motivated their participation, 95 % reporting that it motivated them to invest more effort, and 90 % reporting that it motivated them to regulate their learning. The main contribution of their responses lies in providing insights into the reasons underlying the discovered effects of personalized gamification on student motivational, behavioral and cognitive learning outcomes. Specifically, student responses provide empirical evidence that bridges the inner mechanisms of the above-discussed theories: Personalized gamification enhances student positive emotions and reduces negative emotions, which are antecedents of an improved motivational state (control-value theory). The improved motivational states exert a positive influence on behaviors (social cognitive theory, self-determination theory), and these behaviors, in turn, promote cognitive learning outcomes (gamified learning theory).

4.4. Limitations and directions for future research

This study has a few limitations. First, the present study adopted a static personalization method in which students' player types were determined by the initial player type questionnaire, and no later change was considered. Although interviewees reported that their player types remained relatively stable within the 2-month intervention period, some believed that their player types might change in the long run. Santos et al. (2023) provided preliminary evidence supporting this assumption by analyzing the association of player types in two rounds of data collection over six months. Future studies could consider adopting a dynamic personalized gamification approach that automatically adjusts to changes in player types. Additionally, player type is only one dimension of the personalization criterion, and contextual differences, such as subject matter, may interact with individual differences and collectively affect the effectiveness of personalized gamification. Therefore, future studies could further explore multidimensional personalized gamification methods (Rodrigues et al., 2023).

Second, we did not include measures of students' intrinsic motivation and cognitive engagement in the course in the pre-intervention questionnaire due to the difficulty students might face in reporting detailed information about these factors *before* actually experiencing the course. Additionally, there is an ethical concern regarding the potential stress or discomfort caused by asking students to respond to too many questions, especially considering that they have already answered a player type questionnaire. Although students expressed similar interest in the course when voluntarily signed up for it, caution should be exercised when interpreting the findings regarding questionnaire responses. It would be more rigorous to include an initial measurement in future studies.

Third, the results of this study are restricted to the use of our newly proposed Pentagon player type model. It is important to acknowledge that we drew inspiration from existing player type models, i.e., Bartle's taxonomy (Bartle, 1996), Yee's gamer motivation profile (Yee et al., 2012), BrainHex (Nacke et al., 2014), and Hexad (Tondello et al., 2016). Guided by the principle of parsimony, we synthesized common concepts from previous player type models (Appendix A). Although preliminary exploratory analyses suggest that distinguishing competition from challenge and including autonomy yields promising psychometric properties (Appendix B), future research should conduct direct comparisons between our model and alternative models. This can be achieved by asking the same participants to complete multiple surveys, if feasible. Such a comparison would provide a fairer and more robust test of the relative merits of each model.

Fourth, there is a limitation in using task completion as a measure of engagement, as it may not accurately reflect the level of engagement of a participant. Relating engagement to the rate of completion may not account for differences in conscientiousness among participants. For example, a participant could complete all the tasks with little effort or need for correctness, even if they are disengaged. Another participant may be fully engaged but face challenges in completing a task, resulting in a lower rate of completion. Therefore, relying solely on task completion as a measure of engagement may not fully capture the nuanced and complex nature of engagement in certain contexts. Future studies could consider using alternative methods, such as trace data, to measure student behavioral engagement.

Fifth, this study did not include a non-gamified control group. Future research may consider including a non-gamified control group to assess how the effects of personalized gamification might differ from a non-gamified condition. Incorporating a non-gamified baseline control group may provide a clearer understanding of the unique effects of personalized gamification on motivational, behavioral, and cognitive learning outcomes. The current study was limited to a relatively small student sample which can affect the generalizability of the results. Future studies should gather data from other student samples, and larger sample sizes to enhance the generalizability of the findings.

Finally, in terms of the interview method, interviews were only conducted on participants from the personalized condition considering the main purpose of conducting interviews was to shed light on the underlying mechanism through which personalized gamification influences students' motivational, behavioral and cognitive learning outcomes. Conducting individual interviews from the control group would provide more direct evidence for the differences in students' perceptions between the two conditions, thereby enhancing the interpretation of the causal effect.

5. Conclusions

Personalized gamification based on player types has been touted as a promising means to address the challenges associated with the conventional one-size-fits-all gamification and has received increasing attention. However, several research gaps exist in this area. Firstly, previous player type models have shown conceptual overlap, and the related measurements have reported suboptimal psychometric properties. Secondly, while efforts have been made to model the relationships between player types and students' motivational and behavioral outcomes in gamified learning, only a limited number of studies have examined the effectiveness of personalized gamification using experimental designs. Thirdly, many previous experiments have failed to employ appropriate methods and have reported inconsistent findings: (a) the control group has often only involved a non-gamified approach;

(b) the focus has been on only one or two aspects of motivational, behavioral, and cognitive learning outcomes; (c) the duration of the experiments has been short (around 30 min) or not reported.

To address these research gaps, the current study first proposed a new player type model by synthesizing significant distinctions and commonalities among previous models. Accordingly, an empirically validated player type questionnaire based on this model was introduced in the experiment. The study employed a mixed-method research design to examine the effects of personalized gamification on student learning in terms of motivational (intrinsic motivation), behavioral (task completion rate), and cognitive (cognitive engagement and learning performance) outcomes, comparing it to a well-designed one-size-fits-all gamification approach over eight weeks. The findings supported the effectiveness of personalized gamification in promoting student learning and shed light on the underlying mechanism through which personalization induces positive emotions, which increases intrinsic motivation, leads to positive behavioral changes, and ultimately enhances cognitive learning outcomes.

Theoretically, this study integrated insights from the control-value theory, social cognitive theory, self-determination theory, and gamified learning theory to connect the constructs of emotion, motivation, behavior, and cognition. In practical terms, the study provides implications for researchers to implement personalized gamification designs in online learning.

Supplementary data to this article can be found online at https://doi.org/10.1016/j.lindif.2024.102470.

Ethics statement

This study was approved by the Human Research Ethics Committee of the authors' institution. Informed consent was obtained from all subjects involved in the study.

CRediT authorship contribution statement

Ya Xiao: Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Khe Foon Hew:** Writing – review & editing, Supervision, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no conflicts of interest.

Acknowledgements

The authors would like to thank Mr Xueliang Chen at the University of Hong Kong for his valuable assistance in analyzing the interview data of this study.

Appendix A. Comparison of the Pentagon player type model with previous player type models

Pentagon player types and motivational stimulus	Bartle's taxonomy	Yee's model	BrainHex	Hexad
Challenger Self-advancement through completing goals or overcoming challenges	**Explorer** Tackling challenging problems (puzzle-solving)	**Achiever** Overcoming challenges	**Achiever, Mastermind, Conqueror** *Achiever*: accomplishing goals; *Mastermind*: tackling challenging problems (puzzle-solving); *Conqueror*: struggling against adversity	**Achiever** Completing challenging tasks
Competitor Competition with others	/	**Achiever** Excelling over others	**Conqueror** Defeating other players	/
Team Player Working together to accomplish shared objectives	**Socializer** Forming connections with other players	**Socializer** Engaging and building connections with other players	**Socializer** Building connections with other players	**Philanthropist** Cooperative or collaborative relationships with others
Explorer New things (e.g., surprises) and new information	**Explorer** Curiosity towards new information	**Immersion** Exploration in the virtual world	**Seeker** Exploring new things	**Free Spirit** Curiosity towards new experience
Autonomy Seeker Autonomy in performing tasks	/	/	/	**Free Spirit** Autonomy in performing tasks (self-expression)

Note. Three essential concepts identified across previous player type models are challenge, teamwork, and exploration. Challenge represents the attribute of overcoming obstacles, which is derived from Bartle's Explorer (Bartle, 1996), Yee's Achiever (Yee et al., 2012), BrainHex's Achiever, Conqueror, and Mastermind (Nacke et al., 2014), as well as Hexad's Achiever (Tondello et al., 2019). Teamwork represents the attribute of working together to achieve shared objectives, which is derived from Bartle's Socializer, Yee's Socializer, BrainHex's Socializer, and Hexad's Philanthropist. Exploration represents the attribute of discovering new things, which is derived from Bartle's Explorer, BrainHex's Seeker, Yee's Immersion, and Hexad's Free Spirit.
In addition, Autonomy Seekers represents the attribute of seeking autonomy in performing tasks. This is delineated from Hexad's Free Spirit (Tondello et al., 2019). Competitor represents the attribute of competing with others. This is delineated from previous conceptualizations, i.e., Yee's Achiever (Yee et al., 2012) and BrainHex's Conqueror (Nacke et al., 2014).

Appendix B. Fit statistics of the hypothesized model and competing models

Model	CFI	TLI	RMSEA (90 % CI)	SRMR
Hypothesized model	0.91	0.90	0.07 [0.06, 0.07]	0.07
BrainHex & Yee models (merging Competitor and Challenger)	0.80	0.78	0.11 [0.10, 0.11]	0.09
Hexad model (merging Explorer and Autonomy Seeker)	0.84	0.82	0.10 [0.09, 0.10]	0.09

Note. This Table provides a preliminary exploration of how incorporating practices from previous models may change the results, rather than presenting a direct comparison.
For example, in BrainHex's "Conqueror" and Yee's "Achiever", the dimensions of Challenger and Competitor are not differentiated, but combined together. Therefore, to provide an estimate of how our hypothesized model, which differentiates Challenger and Competitor, might compare to the BrainHex and Yee models, we merge these two dimensions into one factor. However, this is merely an approximate evaluation. To provide a fairer evaluation of each model's strengths and weaknesses, future studies should have the same participants complete multiple surveys, if feasible.

CFI: comparative fit index, TLI: Tucker-Lewis index, RMSEA: root mean square error of approximation, 90 % CI: 90 % confidence interval, SRMR: standardized root mean square residual.

Appendix C. The Pentagon player type questionnaire

Challenger

1. I like overcoming challenges.
2. I like mastering new skills and to improve myself.
3. I like completing every goal I set.
4. It is important for me to keep challenging myself.
5. It is important for me to overcome difficulties to achieve success.

Competitor

6. I like want to do better than a strong opponent.
7. I like to be ranked in the top.
8. It is important for me to have competition.
9. I am a competitive person.
10. Competition brings the best out of me.

Team Player

11. I enjoy group activities.
12. I like cooperating with others.
13. It is important for me to be part of a team.
14. I like contributing to the team.

Explorer

15. I like discovering the hidden clues (e.g., implied learning points in the slides).
16. I am very curious about new things.
17. I like to have exploratory tasks (e.g., exploring external links related to the course topic).
18. (R) I don't like exploring too much by myself.
19. I like unexpected things.

Autonomy Seeker

20. I like to plan my learning rather than have the instructor plan for me.
21. I like to decide what to do rather than have the instructor tell me what to do.
22. I enjoy working out an effective solution to a problem on my own.
23. Deciding for myself what to do is important to me.
24. I enjoy having a voice in deciding assignments.

References

Abeysekera, L., & Dawson, P. (2015). Motivation and cognitive load in the flipped classroom: Definition, rationale and a call for research. *Higher Education Research and Development, 34*(1), 1–14. https://doi.org/10.1080/07294360.2014.934336

Armstrong, M. B., & Landers, R. N. (2017). An evaluation of gamified training: Using narrative to improve reactions and learning. *Simulation & Gaming, 48*(4), 513–538. https://doi.org/10.1177/1046878117703749

Artino, A. R., La Rochelle, J. S., Dezee, K. J., & Gehlbach, H. (2014). Developing questionnaires for educational research: AMEE guide no. 87. *Medical Teacher, 36*(6), 463–474. https://doi.org/10.3109/0142159X.2014.889814

Awais, M., Habiba, U., Khalid, H., Shoaib, M., & Arshad, S. (2019). An adaptive feedback system to improve student performance based on collaborative behavior. *IEEE Access, 7*, 107171–107178. https://doi.org/10.1109/ACCESS.2019.2931565

Bai, S., Hew, K. F., & Huang, B. (2020). Does gamification improve student learning outcome? Evidence from a meta-analysis and synthesis of qualitative data in educational contexts. *Educational Research Review, 30*, Article 100322. https://doi.org/10.1016/j.edurev.2020.100322

Bai, S., Hew, K. F., Sailer, M., & Jia, C. (2021). From top to bottom: How positions on different types of leaderboard may affect fully online student learning performance, intrinsic motivation, and course engagement. *Computers & Education, 173*, Article 104297. https://doi.org/10.1016/j.compedu.2021.104297

Bandura, A. (1977). Self-efficacy: Toward a unifying theory of behavioral change. *Psychological Review, 84*, 191–215. https://doi.org/10.1037/0033-295X.84.2.191

Bartle, R. (1996). Hearts, clubs, diamonds, spades: Players who suit MUDs. *Journal of MUD Research, 1*(1), 19.

Bateman, C., & Boon, R. (2006). *21st century game design (game development series)*. Charles River Media, Inc.

Bateman, C., Lowenhaupt, R., & Nacke, L. E. (2011). Player typology in theory and practice. In *Proceedings of DiGRA 2011* (pp. 1–24). https://hcigames.com/wp-content/uploads/2015/01/Player-Typology-in-Theory-and-Practice.pdf.

Bateman, C., & Nacke, L. E. (2010). The neurobiology of play. In *Proceedings of the international academic conference on the future of game design and technology, Canada* (pp. 1–8).

Bedwell, W. L., Pavlas, D., Heyne, K., Lazzara, E. H., & Salas, E. (2012). Toward a taxonomy linking game attributes to learning: An empirical study. *Simulation & Gaming, 43*(6), 729–760. https://doi.org/10.1177/1046878112439444

Bovermann, K., & Bastiaens, T. (2020). Towards a motivational design? Connecting gamification user types and online learning activities. *Research and Practice in Technology Enhanced Learning, 15*(1), 1–18. https://doi.org/10.1186/s41039-019-0121-4

Broadbent, J. (2017). Comparing online and blended learner's self-regulated learning strategies and academic performance. *The Internet and Higher Education, 33*, 24–32. https://doi.org/10.1016/j.iheduc.2017.01.004

Brown, T. A. (2015). *Confirmatory factor analysis for applied research*. Guilford publications.

Busch, M., Mattheiss, E., Orji, R., Fröhlich, P., Lankes, M., & Tscheligi, M. (2016). Player type models: Towards empirical validation. In *Proceedings of the 2016 CHI conference extended abstracts on human factors in computing systems* (pp. 1835–1841).

Challco, G. C., Isotani, S., & Bittencourt, I. I. (2019). The effects of ontology-based gamification in scripted collaborative learning. In *IEEE 19th international conference on advanced learning technologies (ICALT), Brazil* (pp. 140–144). https://doi.org/10.1109/ICALT.2019.00043

Challco, G. C., Mizoguchi, R., & Isotani, S. (2018). Using ontology and gamification to improve students' participation and motivation in CSCL. In *First international workshop on social, semantic, adaptive and gamification techniques and technologies for distance learning, Brazil* (pp. 174–191). https://doi.org/10.1007/978-3-319-97934-2_11

Cohen, J. (1969). *Statistical Power Analysis for the Behavioral Sciences*. Academic Press, Inc.

Daghestani, L. F., Ibrahim, L. F., Al-Towirgi, R. S., & Salman, H. A. (2020). Adapting gamified learning systems using educational data mining techniques. *Computer Applications in Engineering Education, 28*(3), 568–589. https://doi.org/10.1002/cae.22227

Deci, E. L., Eghrari, H., Patrick, B. C., & Leone, D. R. (1994). Facilitating internalization: The self-determination theory perspective. *Journal of Personality, 62*, 119–142. https://doi.org/10.1111/j.1467-6494.1994.tb00797.x

Deci, E. L., & Ryan, R. M. (1985). *Intrinsic motivation and self-determination in human behavior*. New York, NY: Plenum. https://doi.org/10.1007/978-1-4899-2271-7

Deterding, S., Sicart, M., Nacke, L., O'Hara, K., & Dixon, D. (2011). Gamification. Using game-design elements in non-gaming contexts. In D. Tan, B. Begole, & W. A. Kellogg (Eds.), *CHI '11 extended abstracts on human factors in computing systems* (pp. 2425–2428). ACM. https://doi.org/10.1145/1979742.1979575.

Donnermann, M., Lein, M., Messingschlager, T., Riedmann, A., Schaper, P., Steinhaeusser, S., & Lugrin, B. (2021). Social robots and gamification for technology supported learning: An empirical study on engagement and motivation. *Computers in Human Behavior, 121*, Article 106792. https://doi.org/10.1016/j.chb.2021.106792

Fallucchi, F., Nosenzo, D., & Reuben, E. (2020). Measuring preferences for competition with experimentally-validated survey questions. *Journal of Economic Behavior & Organization, 178*, 402–423. https://doi.org/10.1016/j.jebo.2020.07.028

Förster, M., Maur, A., Weiser, C., & Winkel, K. (2022). Pre-class video watching fosters achievement and knowledge retention in a flipped classroom. *Computers & Education, 179*, Article 104399. https://doi.org/10.1016/j.compedu.2021.104399

Garris, R., Ahlers, R., & Driskell, J. E. (2002). Games, motivation, and learning: A research and practice model. *Simulation & Gaming, 33*, 441–467. https://doi.org/10.1177/1046878102238607

Greene, B. A. (2015). Measuring cognitive engagement with self-report scales: Reflections from over 20 years of research. *Educational Psychologist, 50*(1), 14–30. https://doi.org/10.1007/s11187-020-00420-1

Guay, F., Vallerand, R., & Blanchard, C. (2000). On the assessment of situational intrinsic and extrinsic motivation: The Situational Motivation Scale (SIMS). *Motivation and Emotion, 24*(3), 175–213. https://doi.org/10.1023/A:1005614228250

Hamari, J., & Tuunanen, J. (2014). Player types: A meta-synthesis. *Transactions of the Digital Games Research Association, 1*(2), 29–53. https://doi.org/10.26503/todigra.v1i2.13

Hu, L. T., & Bentler, P. M. (1999). Cutoff criteria for fit indexes in covariance structure analysis: Conventional criteria versus new alternatives. *Structural Equation Modeling: A Multidisciplinary Journal, 6*(1), 1–55. https://doi.org/10.1080/10705519909540118

Huang, B. Y., Hew, K. F., & Chung, K. L. (2019). Investigating the effects of gamification-enhanced flipped learning on undergraduate students' behavioral and cognitive engagement. *Interactive Learning Environments, 27*(8), 1106–1126. https://doi.org/10.1080/10494820.2018.1495653

Huberman, M., & Miles, M. (1994). *Qualitative data analysis: An expanded sourcebook*. SAGE.

Hughes, N. G., Flockton, J. R., & Cairns, P. (2023). Growing together: An analysis of measurement transparency across 15 years of player motivation questionnaires. *International Journal of Human-Computer Studies, 169*, Article 102940.

Jurado, F., & Meza, E. R. (2017). An exploratory study in the use of gamer profiles and learning styles to build educational video games. *International Journal of Engineering Education, 33*(2B), 797–806.

Kline, P. (1999). *The handbook of psychological testing* (2nd ed.). London: Routledge.

Klock, A. C. T., Gasparini, I., Pimenta, M. S., & Hamari, J. (2020). Tailored gamification: A review of literature. *International Journal of Human-Computer Studies, 144*, Article 102495. https://doi.org/10.1016/j.ijhcs.2020.102495

Krath, J., Schürmann, L., & von Korfesch, H. F. O. (2021). Revealing the theoretical basis of gamification: A systematic review and analysis of theory in research on gamification, serious games and game-based learning. *Computers in Human Behavior, 125*, Article 106963. https://doi.org/10.1016/j.chb.2021.106963

Kwon, H. Y., & Özpolat, K. (2021). The dark side of narrow gamification: Negative impact of assessment gamification on student perceptions and content knowledge. *INFORMS Transactions on Education, 21*(2), 67–81. https://doi.org/10.1287/ited.2019.0227

Kyewski, E., & Krämer, N. C. (2018). To gamify or not to gamify? An experimental field study of the influence of badges on motivation, activity, and performance in an online learning course. *Computers & Education, 118*, 25–37. https://doi.org/10.1016/j.compedu.2017.11.006

Lai, C. L., & Hwang, G. J. (2016). A self-regulated flipped classroom approach to improving students' learning performance in a mathematics course. *Computers & Education, 100*, 126–140. https://doi.org/10.1016/j.compedu.2016.05.006

Landers, R. N. (2014). Developing a theory of gamified learning. *Simulation & Gaming, 45*(6), 752–768. https://doi.org/10.1177/1046878114563660

Landers, R. N., Auer, E. M., Collmus, A. B., & Armstrong, M. B. (2018). Gamification science, its history and future: Definitions and a research agenda. *Simulation & Gaming, 49*(3), 315–337. https://doi.org/10.1177/1046878118774385

Lavoué, É., Monterrat, B., Desmarais, M., & George, S. (2019). Adaptive gamification for learning environments. *IEEE Transactions on Learning Technologies*, 1–13.

Lee, D. (2019). The convergent, discriminant, and nomological validity of the Depression Anxiety Stress Scales-21 (DASS-21). *Journal of Affective Disorders, 259*, 136–142. https://doi.org/10.1016/j.jad.2019.06.036

Leitão, R., Maguire, M., Turner, S., & Guimarães, L. (2022). A systematic evaluation of game elements effects on students' motivation. *Education and Information Technologies, 27*(1), 1081–1103. https://doi.org/10.1007/s10639-021-10651-8

Liang, M. Z., Chen, P., Knobf, M., Molassiotis, A., Tang, Y., & Ye, Z. J. (2023). Measuring resilience by cognitive diagnosis models and its prediction of 6-month quality of life in Be Resilient to Breast Cancer (BRBC). *Frontiers in Psychiatry, 14*, Article 1102258. https://doi.org/10.3389/fpsyt.2023.1102258

Lincoln, Y. S., & Guba, E. G. (1985). *Naturalistic inquiry*. Beverly Hills, CA: Sage Publications.

Liu, Y., Alexandrova, T., & Nakajima, T. (2011). Gamifying intelligent environments. In *Proceedings of the 2011 International ACM Workshop on Ubiquitous Meta User Interfaces* (pp. 7–12).

Liu, Y., Wang, Q., Jou, M., Wang, B., An, Y., & Li, Z. (2020). Psychometric properties and measurement invariance of the 7-item game addiction scale (GAS) among Chinese college students. *BMC Psychiatry, 20*(1), 1–11. https://doi.org/10.1186/s12888-020-02830-7

Luo, Z. (2022). Gamification for educational purposes: What are the factors contributing to varied effectiveness? *Education and Information Technologies, 27*, 891–915. https://doi.org/10.1007/s10639-021-10642-9

Marczewski, A. (2015). *Even Ninja Monkeys like to play*. London: Blurb Inc.

Monterrat, B., Lavoué, É., & George, S. (2017). Adaptation of gaming features for motivating learners. *Simulation & Gaming, 48*(5), 625–656. https://doi.org/10.1177/1046878117712632

Mora, A., Tondello, G. F., Nacke, L. E., & Arnedo-Moreno, J. (2018). Effect of personalized gameful design on student engagement. In *IEEE global engineering education conference (EDUCON), Spain* (pp. 1925–1933). https://doi.org/10.1109/EDUCON.2018.8363471

Nacke, L. E., Bateman, C., & Mandryk, R. L. (2014). BrainHex: A neurobiological gamer typology survey. *Entertainment Computing, 5*(1), 55–62. https://doi.org/10.1016/j.entcom.2013.06.002

Neubauer, A. B., Kramer, A. C., & Schmiedek, F. (2022). Assessing domain-general need fulfillment in children and adults: Introducing the General Need Satisfaction and Frustration scale. *Psychological Assessment, 34*(11), 1022–1035. https://doi.org/10.1037/pas0001169

Oliveira, W., Hamari, J., Shi, L., Toda, A. M., Rodrigues, L., Palomino, T. P., & Isotani, S. (2023). Tailored gamification in education: A literature review and future agenda. *Education and Information Technologies, 28*, 373–406. https://doi.org/10.1007/s10639-022-11122-4

Oliveira, W., Toda, A., Toledo, P., Shi, L., Vassileva, J., Bittencourt, I. I., & Vassileva, J. (2020). Does tailoring gamified educational systems matter? The impact on students' flow experience. In *Proceedings of the 53rd Hawaii international conference on system sciences, US* (pp. 1226–1235). https://hdl.handle.net/10125/63891.

Ooge, J., De Croon, R., Verbert, K., & Vanden Abeele, V. (2020, November). Tailoring gamification for adolescents: A validation study of Big Five and Hexad in Dutch. In *Proceedings of the annual symposium on computer-human interaction in play* (pp. 206–218).

Pekrun, R. (2006). The control-value theory of achievement emotions: Assumptions, corollaries, and implications for educational research and practice. *Educational Psychology Review, 18*, 315–341. https://doi.org/10.1007/s10648-006-9029-9

Pekrun, R., Marsh, H. W., Elliot, A. J., Stockinger, K., Perry, R. P., Vogl, E., … Vispoel, W. P. (2023). A three-dimensional taxonomy of achievement emotions. *Journal of Personality and Social Psychology, 124*(1), 145–178. https://doi.org/10.1037/pspp0000448

Pessoa, M., Lima, M., Pires, F., Haydar, G., Melo, R., Rodrigues, L., Oliveira, D., et al. (2024). A journey to identify users' classification strategies to customize game-based and gamified learning environments. *IEEE Transactions on Learning Technologies, 17*, 527–541. https://doi.org/10.1109/TLT.2023.3317396

Pintrich, P. R., Smith, D. A., García, T., & McKeachie, W. J. (1993). Reliability and predictive validity of the Motivated Strategies for Learning Questionnaire (MSLQ). *Educational and Psychological Measurement, 53*(3), 801–813. https://doi.org/10.1177/0013164493053003024

Pintrich, P. R., Smith, D. A. F., Garcia, T., & McKeachie, W. J. (1991). *A manual for the use of the Motivated Strategies for Learning Questionnaire (MSLQ)*. Ann Arbor, Michigan: The University of Michigan.

R Core Team. (2022). *R: A language and environment for statistical computing*. Vienna, Austria: R Foundation for Statistical Computing. https://www.R-project.org/.

Rattray, J., & Jones, M. C. (2007). Essential elements of questionnaire design and development. *Journal of Clinical Nursing, 16*(2), 234–243. https://doi.org/10.1111/j.1365-2702.2006.01573.x

Revuelta, J., Halty, L., & Ximénez, C. (2018). Validation of a questionnaire for personality profiling using cognitive diagnostic modeling. *The Spanish Journal of Psychology, 21*, E63. https://doi.org/10.1017/sjp.2018.62

Rodrigues, L., Palomino, P. T., Toda, A. M., Klock, A. C., Pessoa, M., Pereira, F. D., … Isotani, S. (2023). How personalization affects motivation in gamified review assessments. *International Journal of Artificial Intelligence in Education*, 1–38. https://doi.org/10.1007/s40593-022-00326-x

Rosseel, Y. (2012). lavaan: An R package for structural equation modeling. *Journal of Statistical Software, 48*(2), 1–36. https://doi.org/10.18637/jss.v048.i0

Ryan, R. M., & Deci, E. L. (2000). Self-determination theory and the facilitation of intrinsic motivation, social development, and well-being. *The American Psychologist, 55*(1), 68–78. https://doi.org/10.1037/0003-066X.55.1.68

Sailer, M., Hense, J. U., Mayr, S. K., & Mandl, H. (2017). How gamification motivates: An experimental study of the effects of specific game design elements on psychological need satisfaction. *Computers in Human Behavior, 69*, 371–380. https://doi.org/10.1016/j.chb.2016.12.033

Sailer, M., & Homner, L. (2020). The gamification of learning: A meta-analysis. *Educational Psychology Review, 32*(1), 77–112. https://doi.org/10.1007/s10648-019-09498-w

Sanchez, D. R., Langer, M., & Kaur, R. (2020). Gamification in the classroom: Examining the impact of gamified quizzes on student learning. *Computers & Education, 144*, Article 103666. https://doi.org/10.1016/j.compedu.2019.103666

Santos, A. C. G., Oliveira, W., Altmeyer, M., Hamari, J., & Isotani, S. (2022). Psychometric investigation of the gamification Hexad user types scale in Brazilian Portuguese. *Scientific Reports, 12*(1), 4920. https://doi.org/10.1038/s41598-022-08820-x

Santos, A. C. G., Oliveira, W., Hamari, J., Joaquim, S., & Isotani, S. (2023). The consistency of gamification user types: A study on the change of preferences over time. In *Proceedings of the ACM on human-computer interaction, 7, CHI PLAY* (pp. 1253–1281). https://doi.org/10.1145/3611068

Santos, A. C. G., Oliveira, W., Hamari, J., Rodrigues, L., Toda, A. M., Palomino, P. T., & Isotani, S. (2021). The relationship between user types and gamification designs. *User Modeling and User-Adapted Interaction, 31*(5), 907–940. https://doi.org/10.1007/s11257-021-09300-z

Schöbel, S. M., Janson, A., & Leimeister, J. M. (2023). Gamifying online training in management education to support emotional engagement and problem-solving skills. *Journal of Management Education, 47*(2), 166–203. https://doi.org/10.1177/10525629221123287

Shaw, R. S. (2013). The relationships among group size, participation, and performance of programming language learning supported with online forums. *Computers & Education, 62*, 196–207. https://doi.org/10.1016/j.compedu.2012.11.001

Shi, Q., Ma, W., Robitzsch, A., Sorrel, M. A., & Man, K. (2021). Cognitively diagnostic analysis using the G-DINA model in R. *Psych, 3*(4), 812–835. https://doi.org/10.3390/psych3040052

Talavera-Mendoza, F., Mamani, A. R., & Deza, A. (2021). Analysis of player user experience with learning objects in a gamified educational platform: Mathematical case. In *2021 5th international conference on education and multimedia technology (ICEMT) (ICEMT), Japan* (pp. 253–259). https://doi.org/10.1145/3481056.3481110

Templin, J. L., & Henson, R. A. (2006). Measurement of psychological disorders using cognitive diagnosis models. *Psychological Methods, 11*(3), 287–305. https://doi.org/10.1037/1082-989X.11.3.287

Tondello, F. G., Valtchanov, D., Reetz, A., Wehbe, R. R., Orji, R., & Nacke, L. E. (2018). Towards a trait model of video game preferences. *International Journal of Human-Computer Interaction, 34*(8), 732–748. https://doi.org/10.1080/10447318.2018.1461765

Tondello, G. F. (2019). *Dynamic personalization of gameful interactive systems* (Unpublished doctoral dissertation). The University of Waterloo.

Tondello, G. F., Mora, A., Marczewski, A., & Nacke, L. E. (2019). Empirical validation of the Gamification User Types Hexad scale in English and Spanish. *International Journal of Human-Computer Studies, 127*, 95–111. https://doi.org/10.1016/j.ijhcs.2018.10.002

Tondello, G. F., Wehbe, R. R., Diamond, L., Busch, M., Marczewski, A., & Nacke, L. E. (2016). The gamification user types hexad scale. In *Proceedings of the 2016 annual symposium on computer-human interaction in play* (pp. 229–243). New York: ACM. https://doi.org/10.1145/2967934.2968082.

Ugur-Erdogmus, F., & Çakır, R. (2022). Effect of gamified mobile applications and the role of player types on the achievement of students. *Journal of Educational Computing Research, 60*(4), 1063–1080. https://doi.org/10.1177/07356331211065679

Van Roy, R., & Zaman, B. (2018). Need-supporting gamification in education: An assessment of motivational effects over time. *Computers & Education, 127*, 283–297. https://doi.org/10.1016/j.compedu.2018.08.018

Xiao, Y., & Hew, K. F. T. (2024). Intangible rewards versus tangible rewards in gamified online learning: Which promotes student intrinsic motivation, behavioural engagement, cognitive engagement and learning performance? *British Journal of Educational Technology, 55*, 297–317. https://doi.org/10.1111/bjet.13361

Xiao, Y., Chen, X., Jin, Y., & Hew, K. F. (2024). Player types in Gamified learning: conceptualization, validation, and profiling. *International Journal of Human–Computer Interaction*, 1–19. https://doi.org/10.1080/10447318.2024.2338664

Yang, C. C. Y., & Ogata, H. (2023). Personalized learning analytics intervention approach for enhancing student learning achievement and behavioral engagement in blended learning. *Education and Information Technologies, 28*, 2509–2528. https://doi.org/10.1007/s10639-022-11291-2

Yee, N., Ducheneaut, N., & Nelson, L. (2012). Online gaming motivations scale: Development and validation. In *Proceedings of the SIGCHI conference on human factors in computing systems* (pp. 2803–2806). https://doi.org/10.1145/2207676.2208681

Zhang, L., Basham, J. D., & Yang, S. (2020). Understanding the implementation of personalized learning: A research synthesis. *Educational Research Review, 31*, Article 100339. https://doi.org/10.1016/j.edurev.2020.100339

Zheng, L., Long, M., Zhong, L., & Gyasi, J. F. (2022). The effectiveness of technology-facilitated personalized learning on learning achievements and learning perceptions: A meta-analysis. *Education and Information Technologies, 27*(8), 11807–11830. https://doi.org/10.1007/s10639-022-11092-7

Zheng, L., Zhong, L., Niu, J., Long, M., & Zhao, J. (2021). Effects of personalized intervention on collaborative knowledge building, group performance, socially shared metacognitive regulation, and cognitive load in computer-supported collaborative learning. *Educational Technology & Society, 24*(3), 174–193. https://www.jstor.org/stable/27032864.

Ya Xiao is a PhD student at the Faculty of Education, the University of Hong Kong. Her research interests lie in technology-assisted learning, gamification, learning engagement and motivation.

Khe Foon Hew is a professor at the Faculty of Education, the University of Hong Kong. His research interests lie in online pedagogy, instructional design, e-learning/blended-learning environments.